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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning and AI Engineer with **3 years** of industry experience owning **ML and GenAI systems end to end** in a regulated financial services setting: researching the approach, designing the architecture, building the software, then deploying and troubleshooting it in production, shipping on **AWS** (EC2 including a GPU instance) with **Docker** and a one-click **Jenkins and GitLab CI/CD pipeline**. Designed the **scoring and ranking algorithm** for an MSc Data Science dissertation with AstraZeneca and completed a **multi-architecture deep learning study**. Led up to 4 interns as **de facto team lead** and was twice recognised with **Best Team Project**. Seeking to build the experimentation, deployment and monitoring foundations behind Primer's ML-powered routing decisions.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production ML & GenAI Systems

- Designed and built **production machine learning and GenAI applications** end to end (**Python FastAPI** microservices, a **ReactJS** frontend, retrieval with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & Llama-3) serving **Compliance, Client Service and Investment**, with a separate **pgvector** database per department, and **agentic tool-calling** where agents call **Python functions** to run **SQL-backed fund comparisons** and generate **PDF reports** on request.
- Owned each system across the **full lifecycle** - researching and testing the approach, designing the architecture, **deploying to real users** and **troubleshooting in production** - and **iterated on what real usage showed** rather than reaching for the most sophisticated method.
- Made the model choice an **engineering trade-off** rather than a default: **self-hosted Llama-3 on a GPU EC2 instance** for the largest department to control cost, stronger **Azure OpenAI** outputs kept for decision makers, and **every prompt change versioned in Git**.

Evaluation, MLOps & Production Reliability

- Kept evaluation **pragmatic and impact-first**: measured what each workflow needed (rating changes matched, analyst hours saved) and **reported results and their limits** plainly to the teams who owned the decision, with quality held by **system-prompt guardrails** rather than an automated eval harness.
- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** (**EC2 including a GPU instance, S3, ECR and Lambda**); packaged the **FastAPI** microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD pipeline**; built reusable pieces rather than one-offs so later applications started from them.
- Operated both **API and batch workloads**: user-facing services on FastAPI microservices, plus a daily rating-email pipeline on **AWS SageMaker** scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM; troubleshot and improved it in production when a source changed its emails.

Data Pipelines, Stakeholder Delivery & Team

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly; developed **Optimized SQL queries, Functions and Procedures**.
- Built a **Python** extraction pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency started sending emails as images.
- Analyzed customer data using **Python Pandas** and **Market Basket Analysis (Apriori Algorithm)** to surface buying patterns for the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.
- Worked directly with **Client Service, Compliance and Investment** stakeholders to scope requirements and turn them into working systems, and talked to the people using them to pressure-test what was built against real usage; led **up to 4 interns** as **de facto team lead** and was recognised with **Best Team Project** twice (2024, 2025).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. I designed the **desirability scoring and ranking algorithm** that orders around 2,100 cancer cell lines by target inclusion and exclusion genes across six evidence layers over DepMap, GEO and HPA data, with a veto for exclusion genes and a calibration step that keeps scores comparable across lineages, plus the preprocessing pipeline and a RAG-based AI assistant; refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Multi-architecture deep learning study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating image descriptions across three synthetic datasets with five seeds per run, reporting SBERT cosine, BLEU and ROUGE-L against BLIP/SigLIP.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator, a car trained using a **Convolutional Neural Network (CNN)** model.

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite